Simulation Study: Sample Mean vs. Sample Median

MT3194 Statistical Inference — Group 11 (Bhumika, Parth, Satyam, Sriharipriya, Vishwabharathi) — simulation component

1 Design

Estimators. Sample mean $\bar{X}_n$ and median M_n, plus three middle-ground estimators: the 10% trimmed mean, the Huber M-estimator ($k = 1.345$, MAD scale; 95% efficient at the normal) and Hodges–Lehmann (median of Walsh averages; ARE $3/\pi$ at the normal). **Models** (i.i.d., centred at $\theta = 0$): $N(0,1)$; Student t_ν, $\nu \in \{1,2,3,4,5,10,30\}$ (t_1 is the Cauchy); Laplace; Tukey scale contamination $(1-\varepsilon)N(0,1) + \varepsilon N(0,3^2)$; asymmetric outliers $(1-\varepsilon)N(0,1) + \varepsilon N(10,1)$, $\varepsilon \in \{.01, .05, .10, .20\}$. $\bar{X}_n$ and M_n run at $n \in \{10, 25, 50, 100, 500, 1000\}$ with $B = 100\,000$ replications; the comparators at $n \in \{25, 100\}$ with $B = 50\,000$. Under the outlier model both estimators are biased for $\theta = 0$, so we compare MSE. **Metrics.** Bias, variance and MSE, with Monte Carlo standard errors from 20 independent batches (relative s.e. of every variance in Table 1: at most 0.6%); n Var against first-order theory ($n\,\mathrm{Var}(\bar{X}_n) = \sigma^2$, $n\,\mathrm{Var}(M_n) \to [4f(m)^2]^{-1}$, Cramér–Rao bound $[nI(\theta)]^{-1}$); and ARE $= \mathrm{Var}(\bar{X}_n)/\mathrm{Var}(M_n)$ against $4\sigma^2 f(m)^2$, where values above 1 favour the median. With infinite variance we report quantiles of |error| and running estimates instead. **Reproducibility.** Master seed 20261110, an independent random stream per cell; code, data and 35 tests are in the repository (`github.com/ParthSarda5/mean-vs-median`); every number below is generated from the saved results.

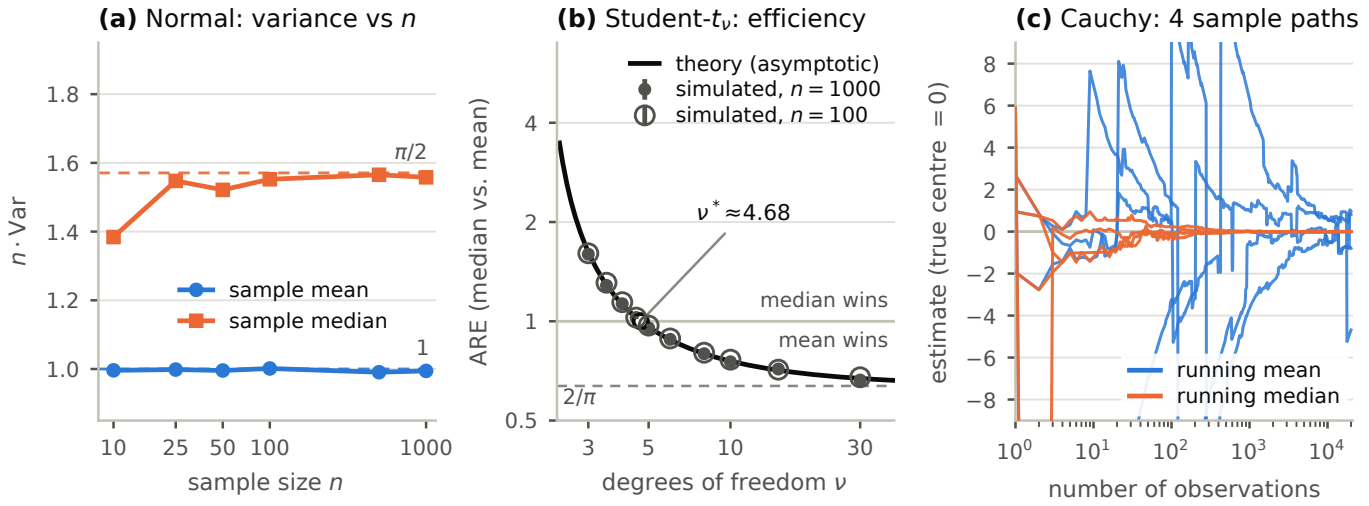


Figure 1: (a) $n\,\mathrm{Var}$ at $N(0,1)$ (95% Monte Carlo bars); dashed: limits 1 and $\pi/2$. (b) Simulated $\mathrm{Var}(\bar{X})/\mathrm{Var}(M)$ for t_ν against the asymptotic curve; the median wins left of ν^*. (c) Running mean and median on four Cauchy streams.

Table 1: Efficiency of the mean and the median ($B = 100\,000$). Bold: median more efficient. ‡CRLB/($n\,\mathrm{Var}$) at $n = 1000$; entries marginally above 1 are Monte Carlo noise. "–": $\mathrm{Var}(\bar{X}_n) = \infty$, so the simulated ratio is meaningless.

	$n \cdot \mathrm{Var}$ at n=1000			ARE $= \mathrm{Var}(\bar{X})/\mathrm{Var}(M)$			eff. vs CRLB‡	
Model	mean	median	med. thy	n=100	n=1000	theory	mean	median
$N(0,1)$	0.99	1.56	1.57	0.645	0.638	0.637	1.01	0.64
t_{10}	1.24	1.64	1.65	0.761	0.758	0.757	0.95	0.72
t_5	1.66	1.73	1.73	0.975	0.960	0.961	0.80	0.77
t_4	2.01	1.78	1.78	**1.140**	**1.127**	**1.125**	0.70	0.78
t_3	2.97	1.86	1.85	**1.603**	**1.598**	**1.621**	0.50	0.81
t_2	∞	2.00	2.00	–	–	∞	0.00	0.83
Cauchy (t_1)	∞	2.45	2.47	–	–	∞	0.00	0.81
Laplace	2.00	1.05	1.00	**1.735**	**1.909**	**2.000**	0.50	0.96
Tukey ε=0.10	1.80	1.81	1.80	1.013	0.998	0.998	0.70	0.70

2 Results

Normal and t models (Fig. 1a,b; Table 1). At $N(0,1)$ the mean attains the bound ($n\,\mathrm{Var} = 0.99$ at $n = 1000$), while $n\,\mathrm{Var}(M_n)$ rises from 1.38 at $n = 10$ (even n averages two central order statistics) to 1.56 at $n = 1000$, giving ARE $= 0.638$ against $2/\pi = 0.637$: the median needs about 57% more data. Across the t family the simulated ARE reproduces $4\sigma^2 f(0)^2$ to within Monte Carlo error (e.g. t_3: 1.598 vs 1.621; t_4: 1.127 vs 1.125). The median overtakes the mean at $\nu^* = 4.68$: ARE $= 1.125$ at $\nu = 4$ but 0.961 at $\nu = 5$. For Laplace data the median is the MLE and the ARE tends to 2, though slowly (1.74 at $n = 100$, 1.91 at $n = 1000$), reflecting the cusp of the density at its median.

Cauchy and infinite variance (Fig. 1c). The sample mean of Cauchy data has the law of a single observation, so its |error| quantiles (50/90/99%) stay at 1.0/6.3/64 for every n (simulated medians: 0.997–1.003) and $n\,\mathrm{Var}(\bar{X}_n)$ explodes (1.7×10^6 at $n = 10$, 1.1×10^9 at $n = 1000$). The median is consistent: its median |error| falls from 0.33 to 0.033, and $n\,\mathrm{Var}(M_n) = 2.45$ matches $\pi^2/4 = 2.47$, i.e. efficiency 0.81 against the bound (theory $8/\pi^2 = 0.81$).

Contamination (Fig. 2d,e). Symmetric scale contamination erodes the mean's advantage steadily: ARE $= 0.83$ at $\varepsilon = .05$, 1.00 at $\varepsilon = .10$ and 1.24 at $\varepsilon = .20$, so a 10% fraction of observations with three times the spread already makes the two estimators equally good. Asymmetric outliers are far worse for the mean: its bias is 10ε (0.50 at $\varepsilon = .05$) against 0.07 for the median, and its MSE is 0.31 versus 0.022 (a factor 14) at $\varepsilon = .05$ and a factor 28 at $\varepsilon = .10$; even $\varepsilon = .01$ triples the mean's n·MSE (2.96 vs. 1.01 when clean). **Sensitivity and breakdown (Fig. 2f,g).** SC_n is the finite-sample influence function: it grows linearly for the mean but is bounded for all others, at ± 1.27 for the median (theory $\pm 1/(2\varphi(0)) = \pm 1.25$) and 1.63–1.67 for the other three.

In the breakdown experiment the mean fails at $k = 1$ (1.0%), the trimmed mean at $k = 11$ (10.9%), Hodges–Lehmann at $k = 30$ (29.7%; theory 29.3%) and the median and Huber at $k = 51$ (50.5%, the maximum possible). **Middle-ground estimators (Table 2).** At the normal the trimmed mean, Huber and HL give up only 6–7% (n·MSE 1.06–1.07 vs. 1.01) whereas the median gives up 55% (1.56). For t_5, t_3 and Tukey contamination they beat *both* the mean and the median (e.g. t_5: 1.33 vs. 1.65/1.70). The median is best only when tails are very heavy or outliers numerous: Laplace (1.17), Cauchy (2.50) and 10% outliers (3.94 vs. Huber 5.68, HL 6.41).

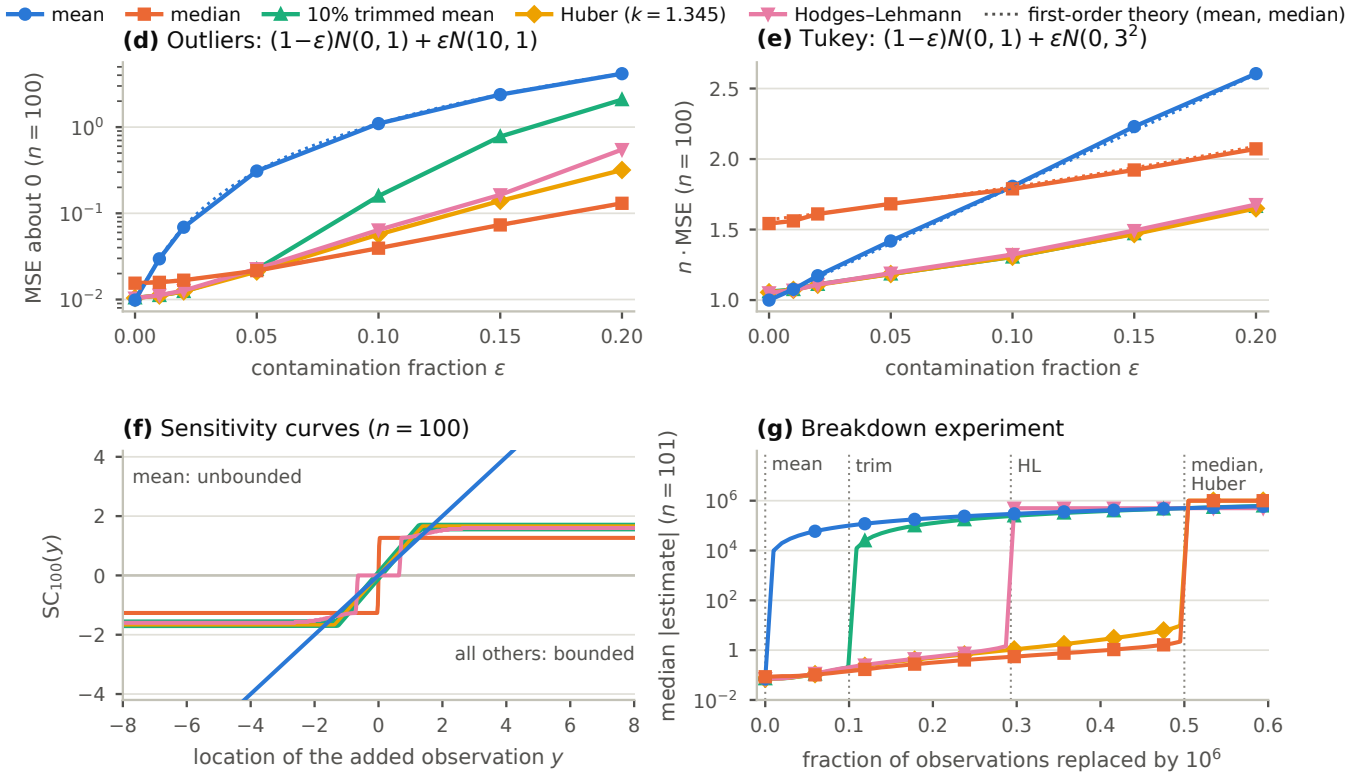


Figure 2: Robustness ($n = 100$ unless stated). (d,e) MSE vs. contamination; dotted: first-order theory (with bias terms 10ε and m_ε in (d)). (f) $\mathrm{SC}_n(y) = n[T(x_1, \ldots, x_{n-1}, y) - T(x_1, \ldots, x_{n-1})]$, $x_i = \Phi^{-1}\left(\frac{i-1/2}{n-1}\right)$. (g) k of $n = 101$ clean points replaced by 10^6; vertical lines: theoretical breakdown points.

Table 2: $n \cdot \mathrm{MSE}$ about $\theta = 0$ at $n = 100$ ($B = 50\,000$; Monte Carlo error about 1%). Lowest in bold (ties at printed precision share it). The mean has no finite MSE for Cauchy data.

Model	mean	median	10% trim	Huber	HL
$N(0,1)$, *theory* ($n \to \infty$)	1.00	1.57	1.06	1.05	1.05
$N(0,1)$	**1.01**	1.56	1.07	1.07	1.06
t_5	1.65	1.70	**1.33**	**1.33**	**1.33**
t_3	2.97	1.84	1.63	**1.60**	**1.60**
Laplace	2.01	**1.17**	1.52	1.46	1.37
Cauchy (t_1)	∞	**2.50**	5.22	3.63	3.44
Tukey ε=0.10	1.81	1.79	**1.31**	**1.31**	1.32
Outliers ε=0.10	109.84	**3.94**	15.86	5.68	6.41

3 Practical guidelines

- **Clean, near-normal data (light tails, no outliers):** use the mean; the median wastes about 36% of the information (ARE ≈ 0.64).
- **Heavy tails (t_ν with $\nu \leq 4$, or infinite variance) or $\gtrsim 10\%$ symmetric contamination:** prefer the median; with Cauchy-type data the mean does not converge at all.
- **Asymmetric gross outliers:** never use the raw mean: 1% contamination triples its n·MSE, and at 5–10% it is 14–28 times worse than the median; prefer the median or a robust compromise.
- **When unsure:** Huber or Hodges–Lehmann cost under 7% at the normal and are never far from the best, and beat both mean and median in the moderate-tail scenarios.
- **Caveats.** Asymptotic ARE is accurate for $n \gtrsim 100$ except for cusp-like densities (Laplace); under skewness the mean and median target different parameters, so comparisons presume symmetry.

Refs. Morris, White & Crowther (2019), *Stat. Med.* 38 (simulation-study design); Tukey (1960); Andrews et al. (1972) *Robust Estimates of Location*; Hodges & Lehmann (1963) *Ann. Math. Stat.*; Huber & Ronchetti (2009) *Robust Statistics*; Hampel et al. (1986) *Robust Statistics*.